

- Create a Vehicle class without any variables and methods
- Create a child class Bus that will inherit all of the variables and methods of the Vehicle class
- Write a Program to create a class by name Students, and initialize attributes like name, age, and grade while creating an object.
- Write a Program to create a valid empty class by name Students, with no properties
- Write a program to create a child class Teacher that will inherit the properties of Parent class Staff
- Write a Python program that checks if one class is a subclass of another.
- Write a Program to create a class and, using the class instance, print all the writable attributes of that object.
- Create a class Teacher with name, age, and salary attributes, where salary must be a private attribute that cannot be accessed outside the class.
- Write a Python program to create an instance of a specified class and display the namespace of the said instance
- Define a Python function student(). Using function attributes display the names of all arguments
- Write a Python function student_data () which will print the id of a student (student_id). If the user passes an argument student_name or student_class the function will print the student name and class.

- Write a simple Python class named Student and display its type. Also, display the `__dict__` attribute keys and the value of the `__module__` attribute of the Student class.
- Write a Python program to create two empty classes, Student and Marks. Now create some instances and check whether they are instances of the said classes or not. Also, check whether the said classes are subclasses of the built-in object class or not.
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- Write a Python class named Student with two attributes: student_name, marks. Modify the attribute values of the said class and print the original and modified values of the said attributes.
- Write a Python class named Student with two attributes student_id, student_name. Add a new attribute student_class and display the entire attribute and their values of the said class. Now remove the student_name attribute and display the entire attribute with values
- Write a Python class named Student with two attributes student_id, student_name. Add a new attribute student_class. Create a function to display the entire attribute and their values in Student class.
- Write a Python class named Student with two instances student1, student2 and assign given values to the said instances attributes. Print all the attributes of student1, student2 instances with their values in the given format
- Write a Python class to find the validity of a string of parentheses, '(', ')', '{', '}', '[' and ']'. These brackets must be closed in the correct order, for example "()" and "()[{}]" are valid but "[)", "({[})" and "{{{" are invalid.
- Write a Python class to find a pair of elements (indices of the two numbers) from a given array whose sum equals a specific target number.
Note: There will be one solution for each input and do not use the same element twice.
Input: numbers= [10,20,10,40,50,60,70], target=50
Output: 3, 4
- Write a Python class to reverse a string word by word.
Input string : 'hello .py'
Expected Output : '.py hello'

- Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle
- Write a Python class which has two methods get_String and print_String. get_String accept a string from the user and print_String print the string in upper case.
- Write a Python class named Rectangle constructed by a length and width and a method which will compute the area of a rectangle
- Create a class Shape that has a method area() and a method perimeter(). Create subclasses Rectangle and Circle that inherit from Shape and override the area() and perimeter() methods as appropriate.
- Create a class Person that has properties name and age. Create a subclass Student that inherits from Person and adds a property major. Create a subclass Teacher that also inherits from Person and adds a property subject. Create instances of each class and print out their properties.
- Create a class Vehicle that has properties make, model, and year. Create a subclass Car that inherits from Vehicle and adds a property num_doors. Create a subclass Motorcycle that also inherits from Vehicle and adds a property num_wheels. Create instances of each class and print out their properties.
- Create a class Animal that has a method speak() that prints out the sound the animal makes. Create subclasses Dog, Cat, and Cow that inherit from Animal and override the speak() method to print out the appropriate sound for each animal. Create instances of each subclass and call their speak() methods.
- Create a class BankAccount that has properties balance and interest_rate. Create a subclass CheckingAccount that inherits from BankAccount and adds a property transaction_fee. Create a subclass SavingsAccount that also inherits from BankAccount and adds a property min_balance. Create instances of each class and print out their properties. Then call a method on each instance that adds interest to the account.
- Create a class Animal that has a method speak() that prints out the sound the animal makes. Create subclasses Dog, Cat, and Cow that inherit from Animal and override the speak() method to print out the appropriate sound for each animal. Then create a list of Animal objects that includes instances of each subclass, and call the speak() method on each object.
- Create a class Shape that has a method area() and a method perimeter(). Create subclasses Rectangle and Circle that inherit from Shape and override the area() and perimeter() methods as appropriate. Then create a list of Shape objects that includes instances of each subclass, and call the area() and perimeter() methods on each object.
- Create a class Person that has properties name and age. Create a subclass Student that inherits from Person and adds a property major. Create a subclass Teacher that also inherits from Person and adds a property subject. Create a list of Person objects that

includes instances of each subclass, and print out their properties. Then call a method on each object that is specific to its subclass (e.g., a method that only exists on Student objects).

- Create a class BankAccount that has properties balance and interest_rate. Create a subclass CheckingAccount that inherits from BankAccount and adds a property transaction_fee. Create a subclass SavingsAccount that also inherits from BankAccount and adds a property min_balance. Create a list of BankAccount objects that includes instances of each subclass, and print out their properties. Then call a method on each object that is specific to its subclass (e.g., a method that only exists on SavingsAccount objects).
- Create a class Vehicle that has properties make, model, and year. Create a subclass Car that inherits from Vehicle and adds a property num_doors. Create a subclass Motorcycle that also inherits from Vehicle and adds a property num_wheels. Create a list of Vehicle objects that includes instances of each subclass, and print out their properties. Then call a method on each object that is specific to its subclass (e.g., a method that only exists on Motorcycle objects)
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